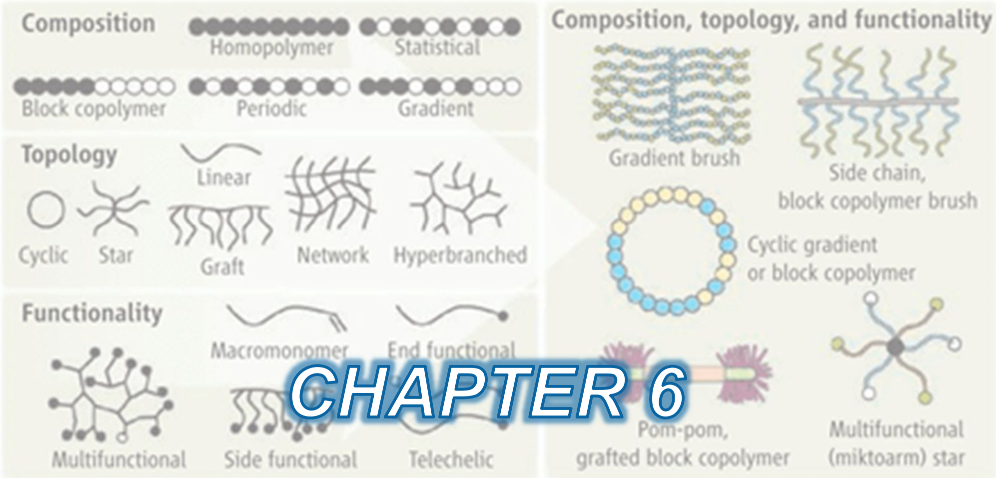
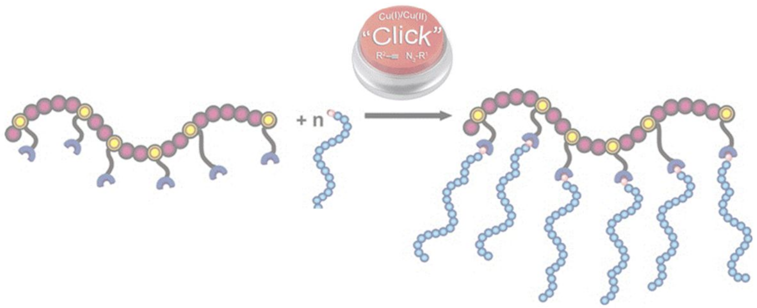
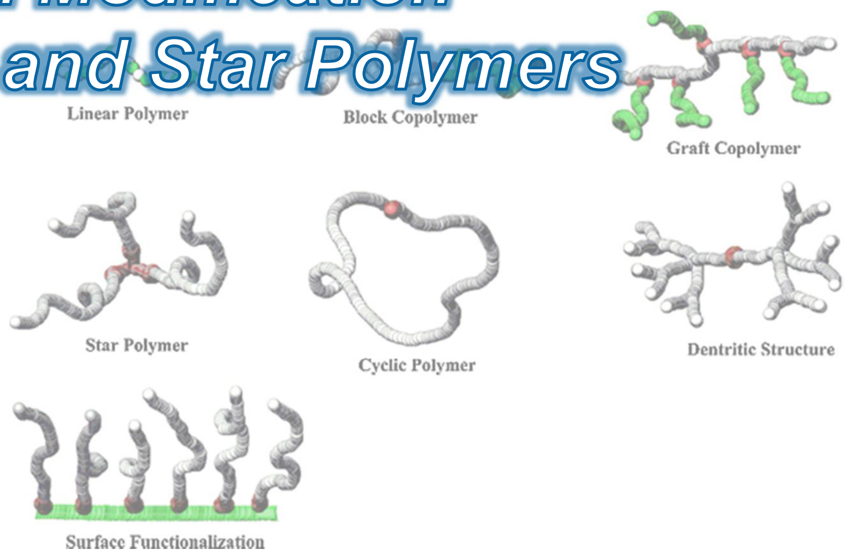


CHAPTER 6

Macromolecular engineering

Postpolymerization Modification and Synthesis of Comb and Star Polymers



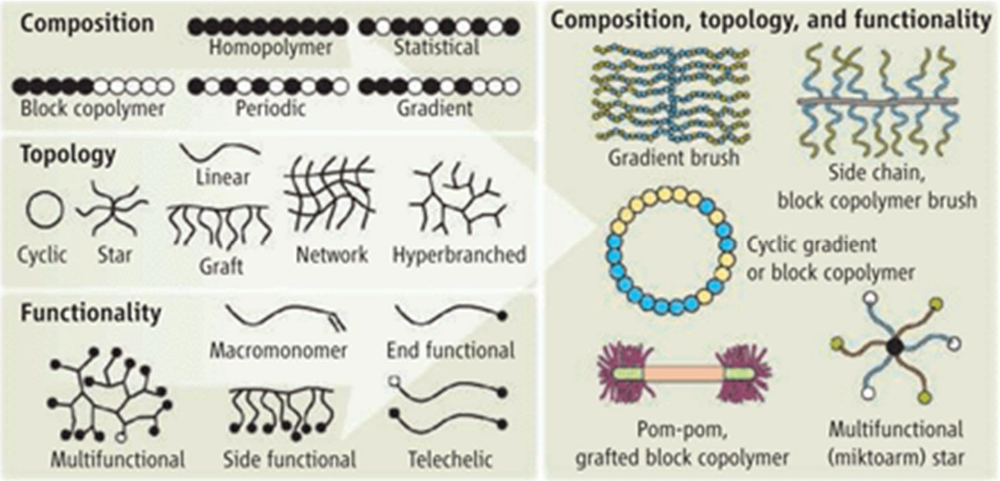


Macromolecular Engineering

= intentional design, synthesis, and control of polymer molecules to achieve *specific structures and properties*

Several aspects of a polymer can be controlled:

- Molecular weight and its distribution
- Chemical composition (*which monomers are incorporated*)
- Sequence of different monomers
- Architecture/topology, (*e.g.*, linear polymers, branched polymers, dendrimers, star polymers, comb polymers, network polymers)
- Chain-end functionality
- Stereochemistry (tacticity)
- Self-assembly into larger nanostructures



Macromolecular Engineering

= intentional design, synthesis, and control of polymer molecules to achieve *specific structures and properties*

Heinrich Heine University Düsseldorf

How is this achieved?

Macromolecular engineering relies on advanced polymerization methods that provide precise **control over chain growth and functionality**, including:

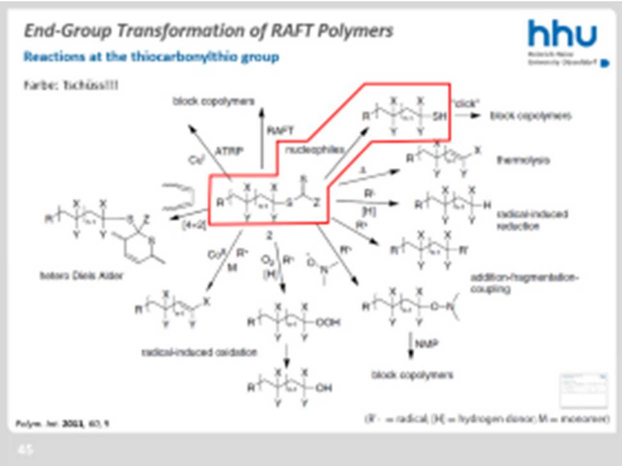
- Living polymerization
- Controlled/pseudo-living radical polymerization (NMP, ATRP, RAFT)
- Ring-opening polymerization
- Use of functional monomers, initiators, CTAs, terminating agents
- Post-polymerization functionalization

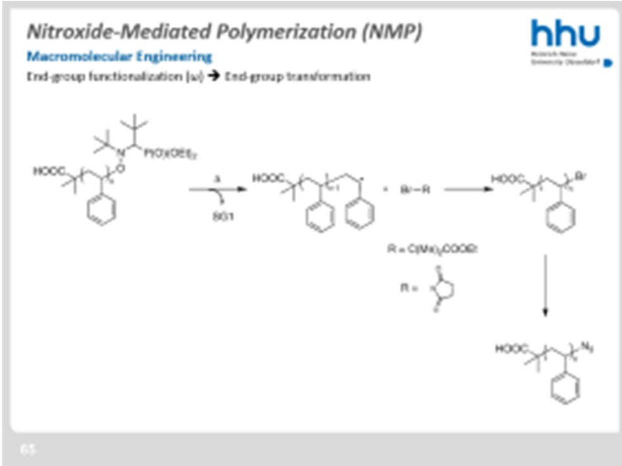
These techniques allow chemists to build polymers almost like molecular "construction sets.“

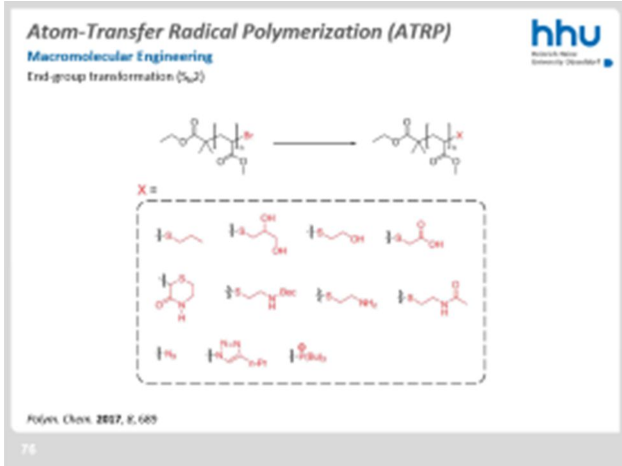
Post-polymerization Modification

End-group modification

see examples from Chapter 1



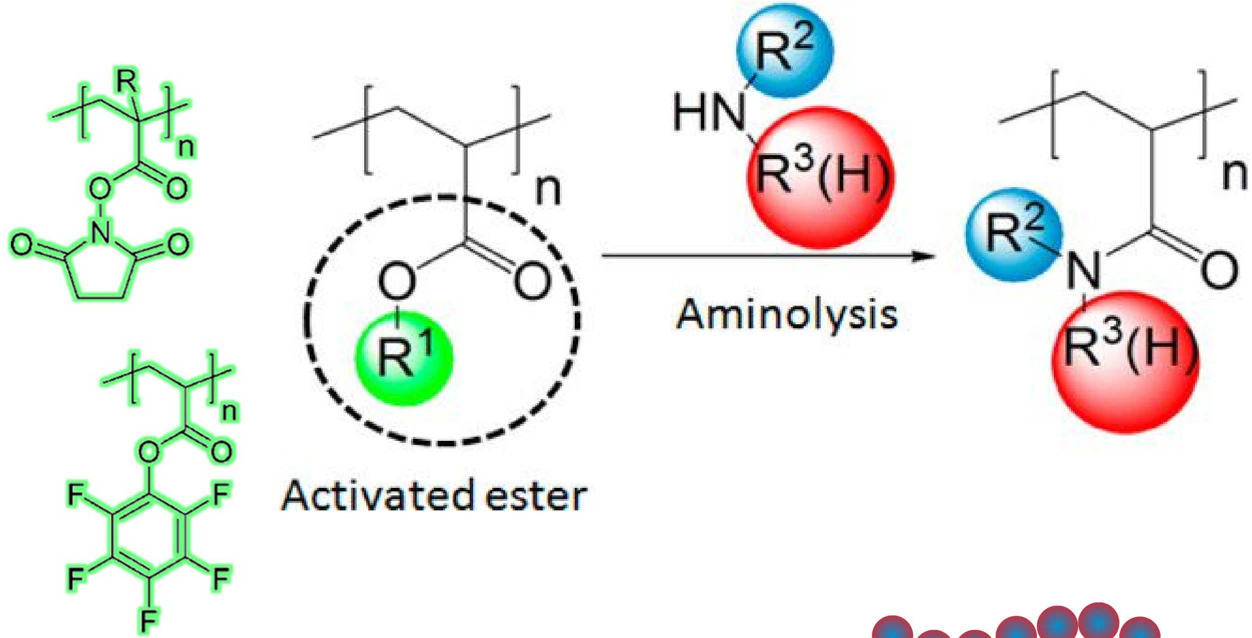


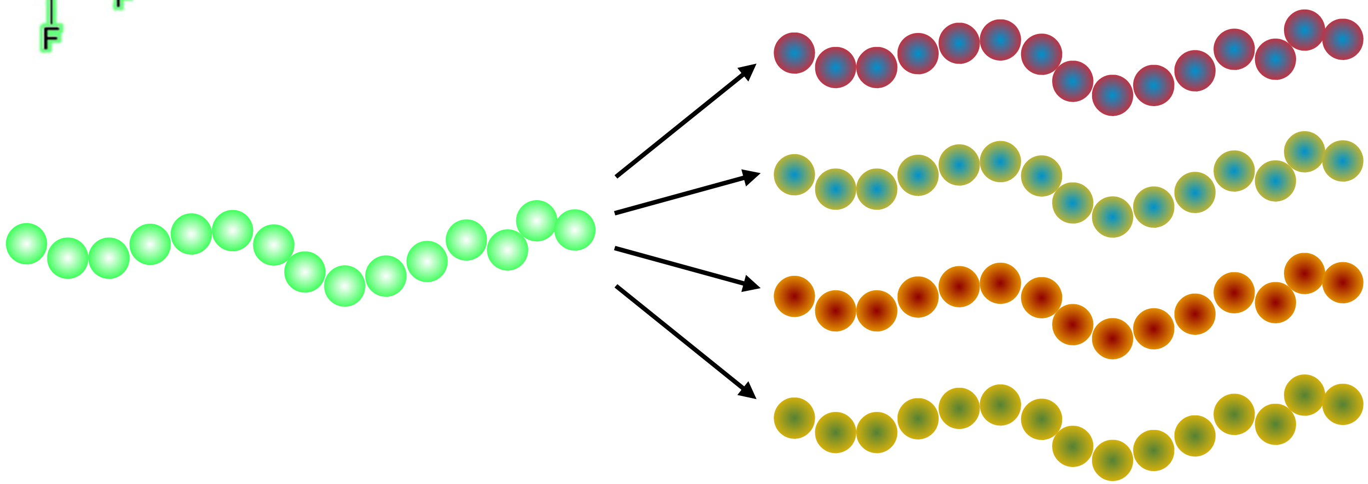


Post-polymerization Modification

Side-chain modification

A prominent example: activated ester-functionalized monomers/polymers





Chem. Rev. **2016**, *116*, 1434

Post-polymerization Modification

Side-chain modification

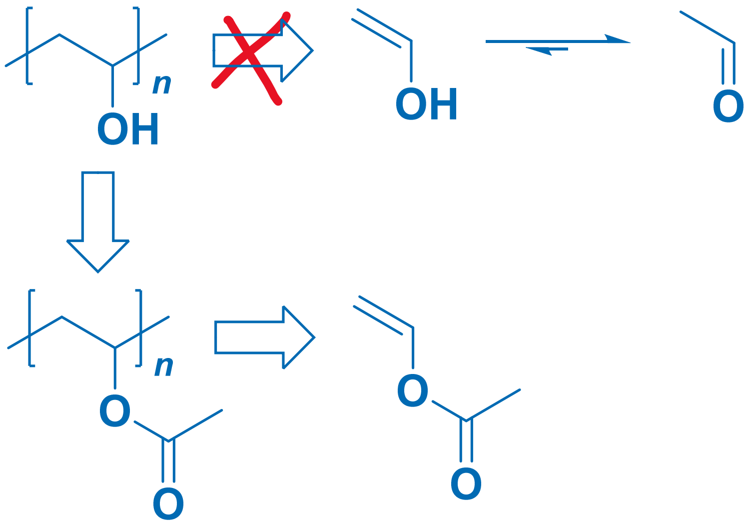
In some cases, it is a necessity to obtain „**indirectly accessible polymers**“ because:

- The corresponding **monomer** is **unstable**, *e.g.*, vinyl alcohol or vinylamine.
- The desired **functional group interferes** with polymerization, *e.g.*, hydroxyls, thiols, or phenols, so protected monomers are used.
- The direct polymerization gives the **wrong architecture**, *e.g.*, aziridine gives branched PEI instead of linear PEI.
- The **polymerization** is **poorly controlled**, so a precursor polymer is synthesized and chemically converted afterward.

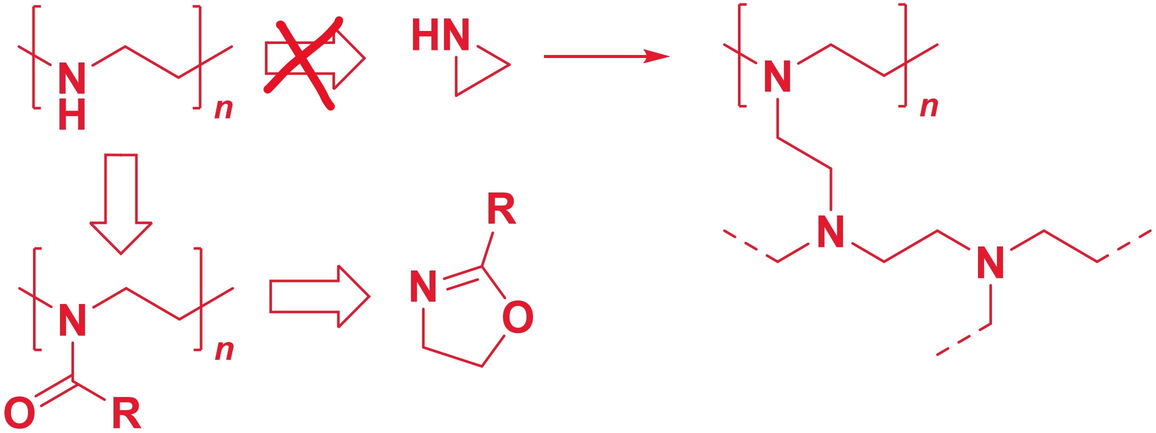
Post-polymerization Modification

Side-chain modification

Ex.: polyvinyl alcohol



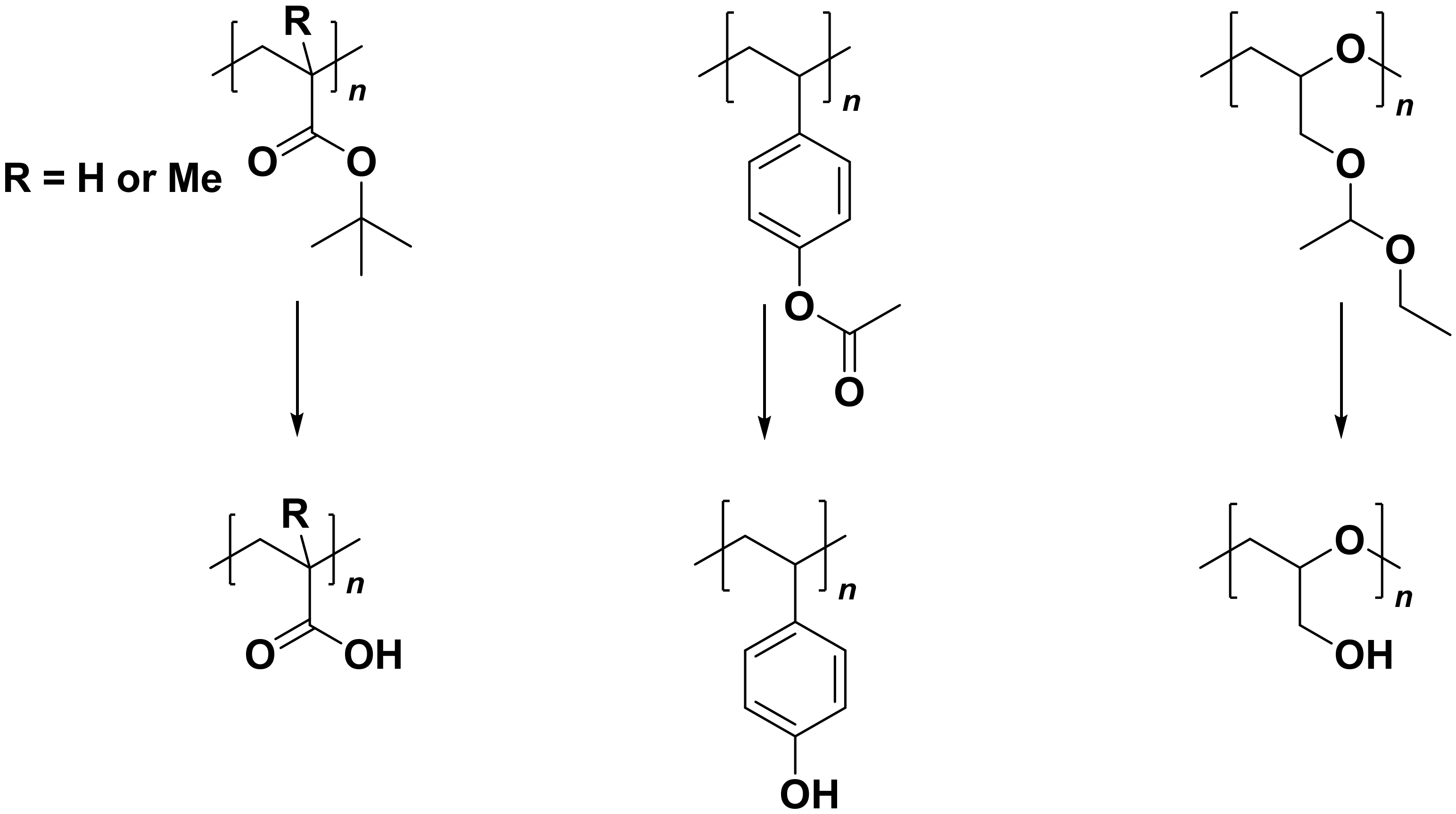
Ex.: linear polyethylene imine



Post-polymerization Modification

Side-chain modification

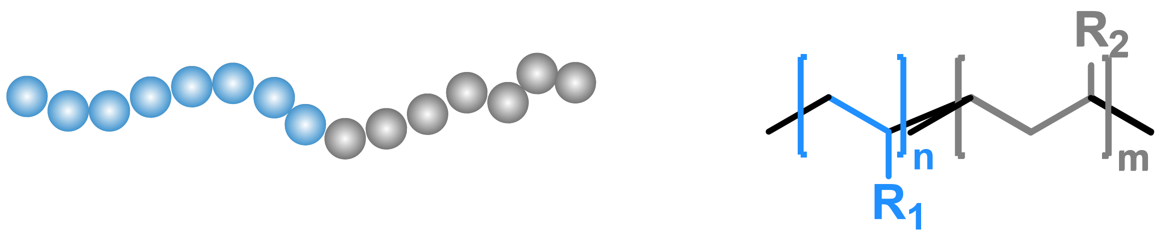
Further examples

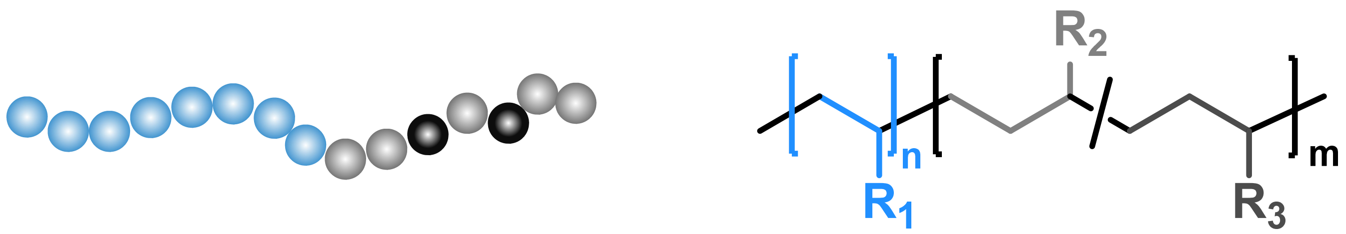


Block Copolymers

Structure and properties

Heinrich Heine
University Düsseldorf



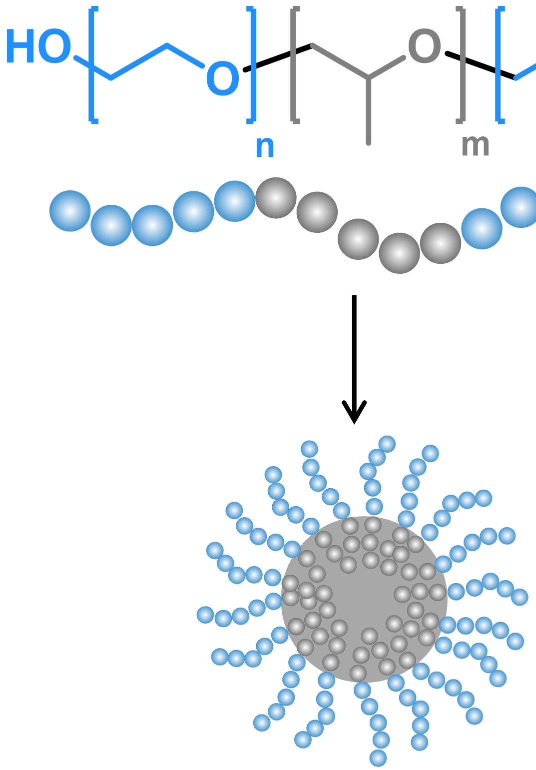


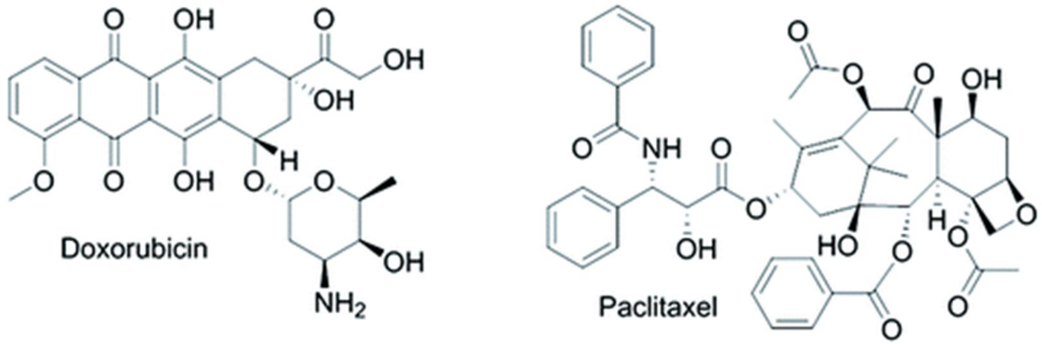
Block Copolymers

Structure and properties

Bringing distinct properties in one macromolecule = at (almost) the same location

Example of Pluronics (poloxamers):
- one **hydrophobic** block (*e.g.,* adhesive to surfaces)
- the other ones bringing **hydrophilicity**





BASF The Chemical Company

Adv. Drug Deliv. Rev. **2002**, *54*, 223; *Polym. Chem.* **2014**, *5*, 3291

J. Mater. Chem. **2012**, *22*, 18596

Block Copolymers

Structure and properties

Bringing distinct properties in one macromolecule = at (almost) the same location

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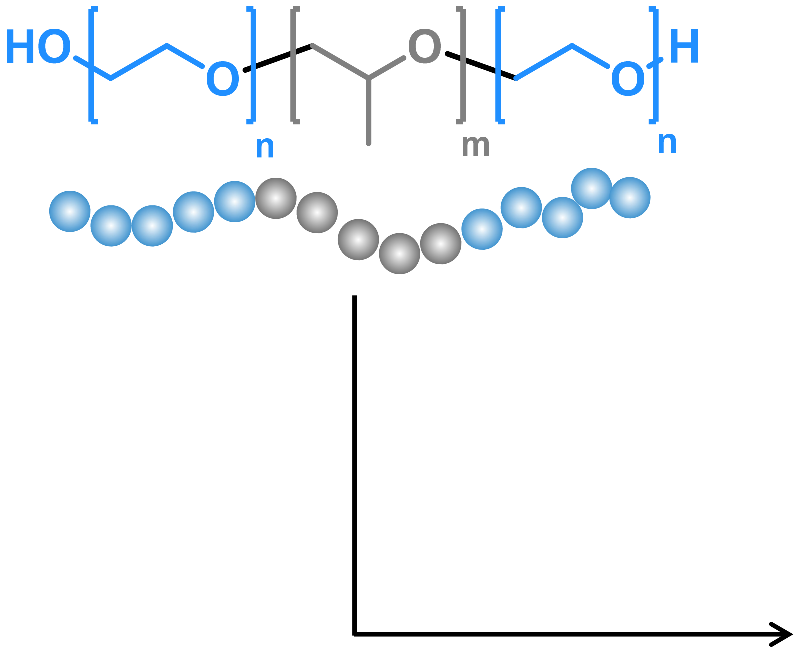
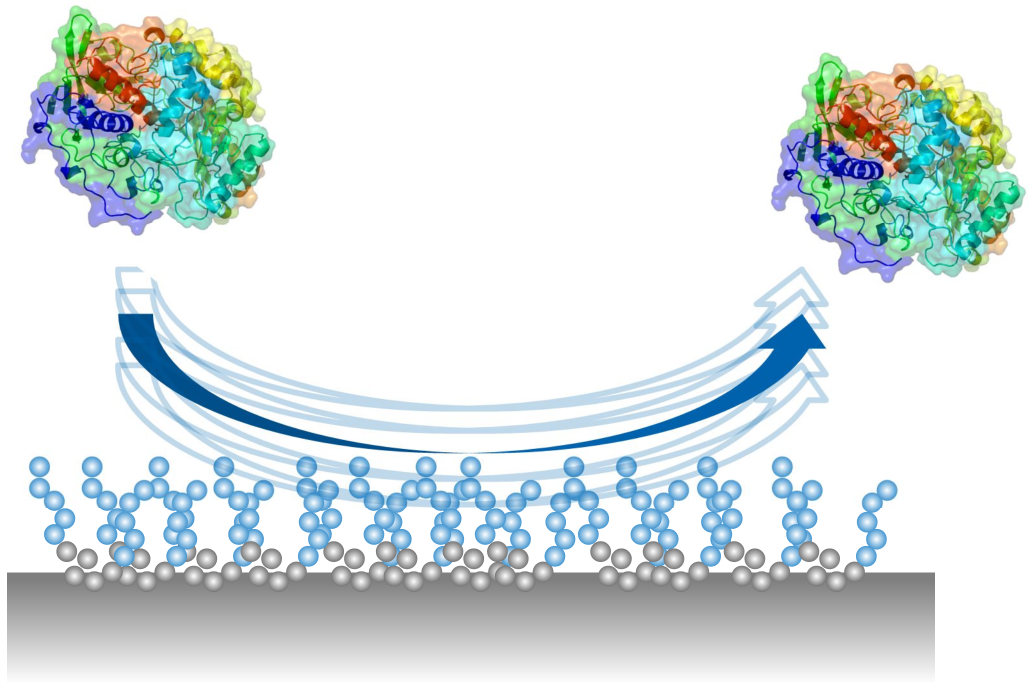


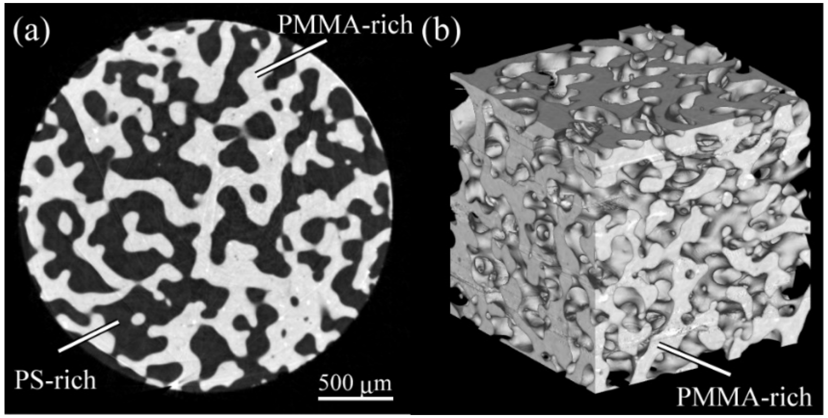


Block Copolymers

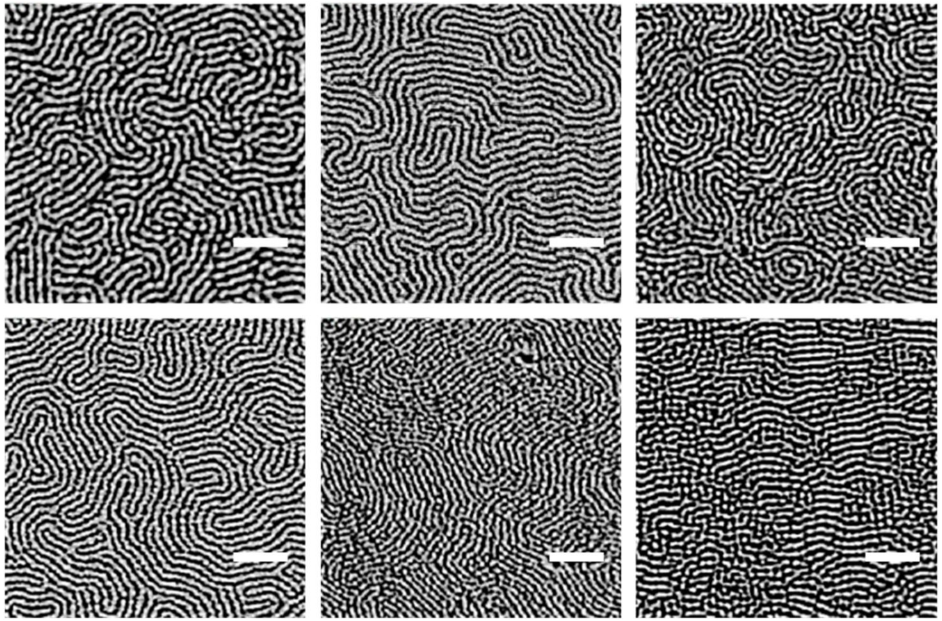
Structure and properties

Most polymers are incompatible: they do NOT mix in the melt.

Imagine what happens when two incompatible polymers are connected to each other…



X-ray phase tomographic image of a PS/PMMA blend



Atomic force micrograph of PS-b-PMMAs
Scale bars 200 nm

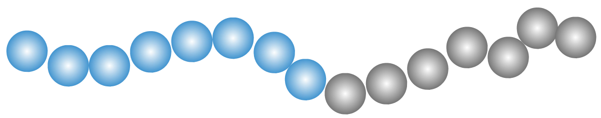
Macromolecules **2004**, *38*, 7197

Delaittre group *Macromol. Rapid Commun.* **2018**, *39*, 1800231

Block Copolymers

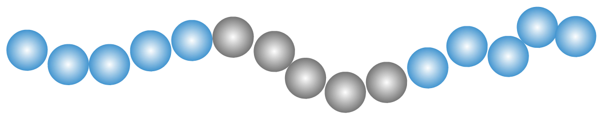
Structure and properties

Linear Diblock Copolymers

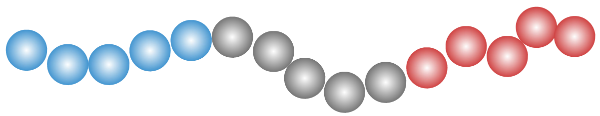


Linear Triblock Copolymers

Symmetrical



Asymmetrical

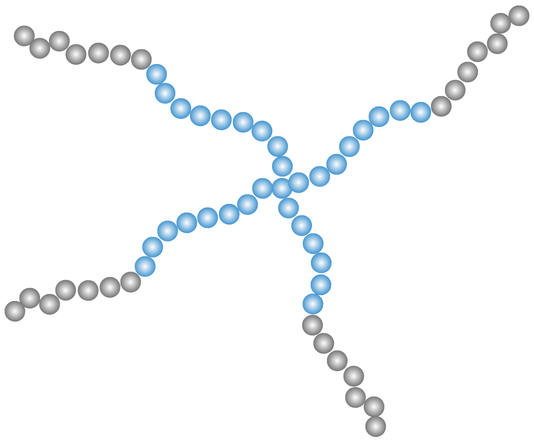


Block Copolymers

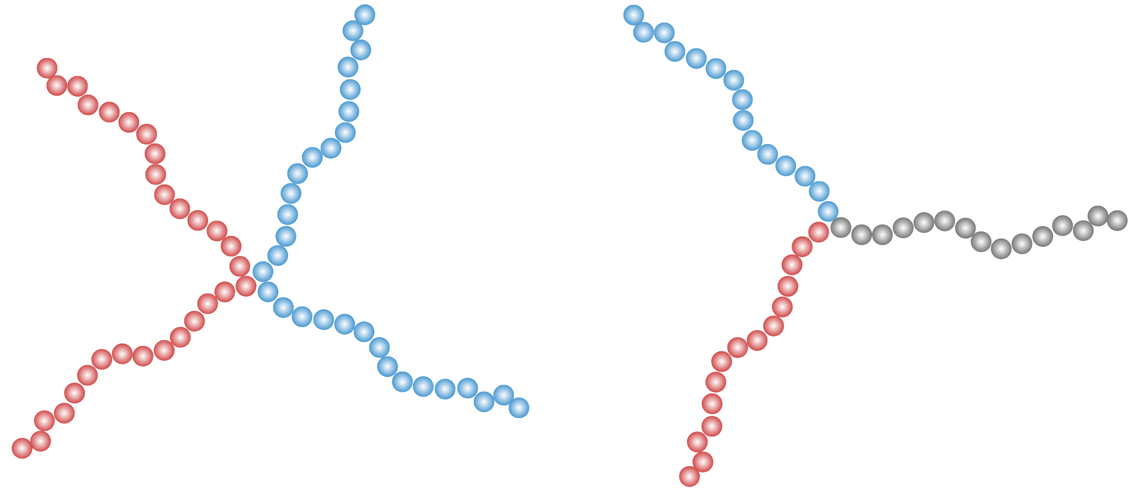
Structure and properties

Star Block Copolymers

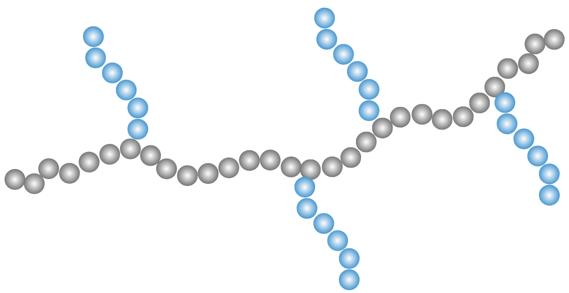
All arms identical



Mikto-arm star copolymers (*μικτός* = mixed)



Graft Copolymers

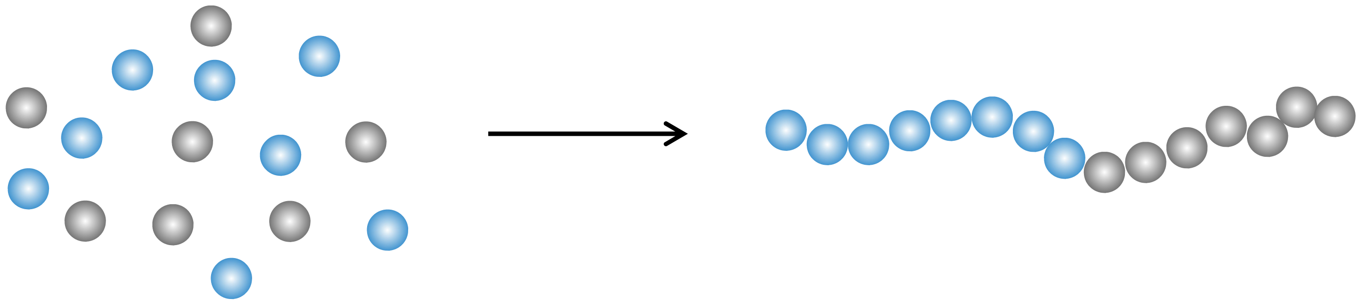


Block Copolymers

Synthetic pathways

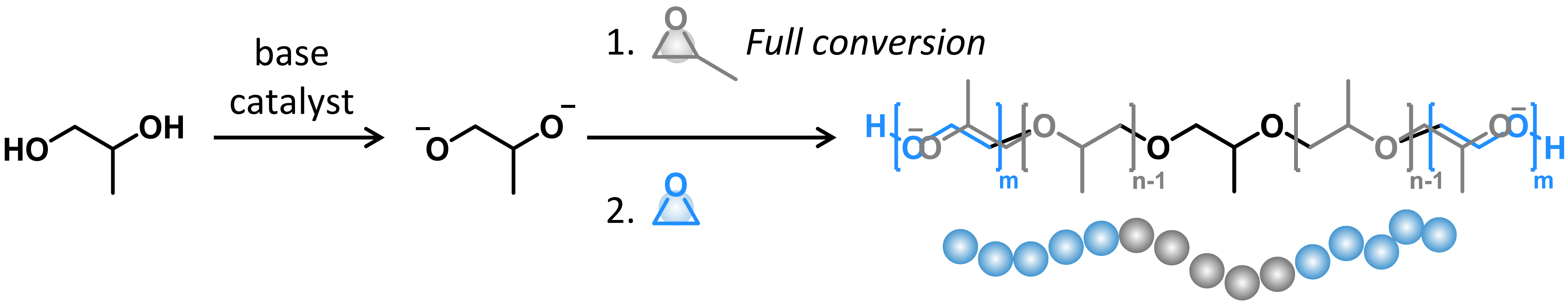
Spontaneous Block Copolymerization ("One Shot")

Favorable reactivity ratios: $r_1 > 1$ and $r_2 > 1$



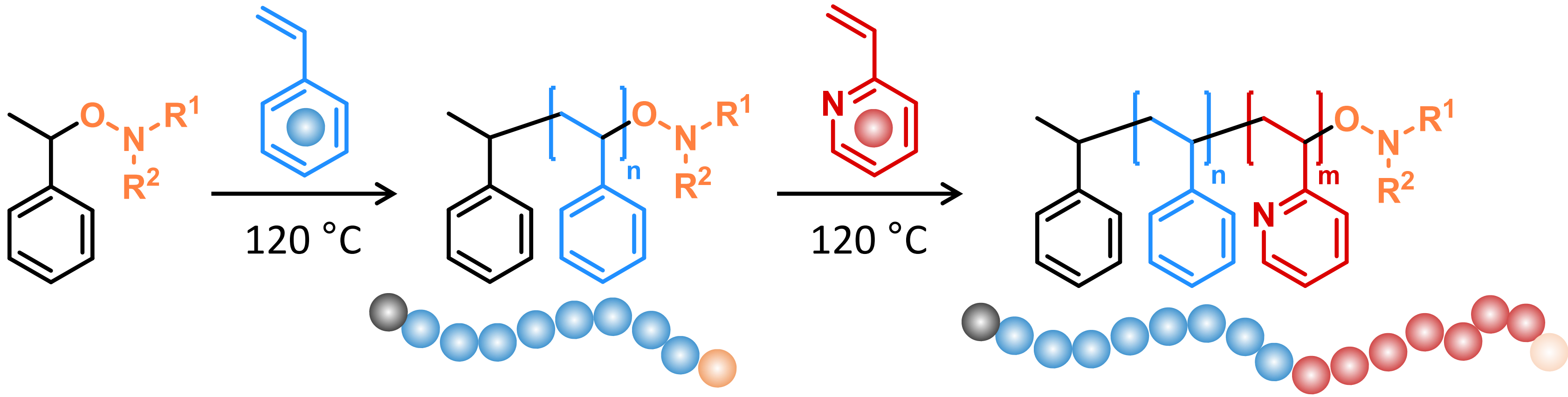
Sequential Addition of Monomers ("One-Pot Two-Stage")

Pluronics



Synthetic pathways

<u>*Macroinitiator*</u> */ Macromolecular Transfer Agent*



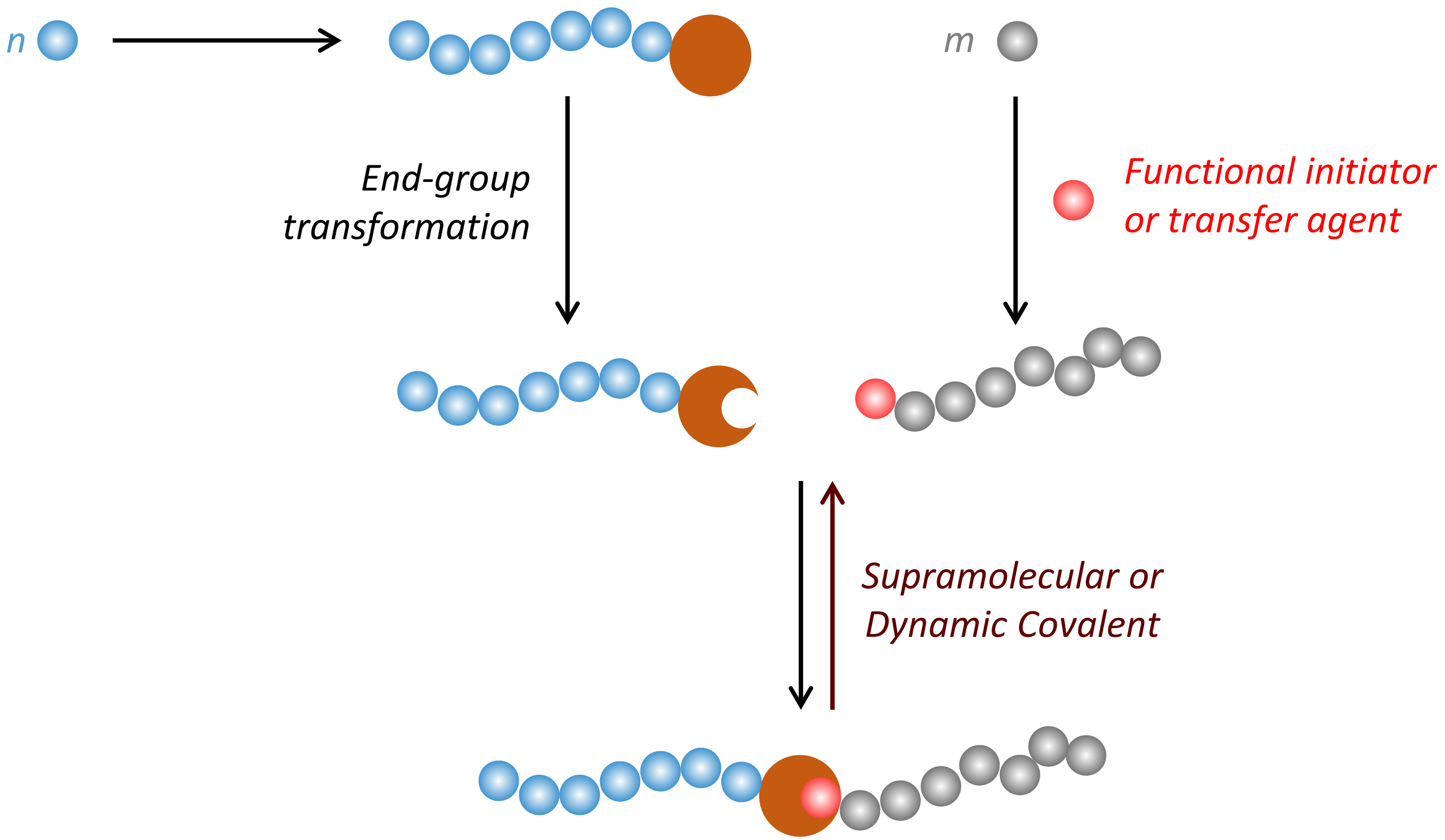
conversion < 70–80 %

Purification by precipitation in methanol

Block Copolymers

Synthetic pathways

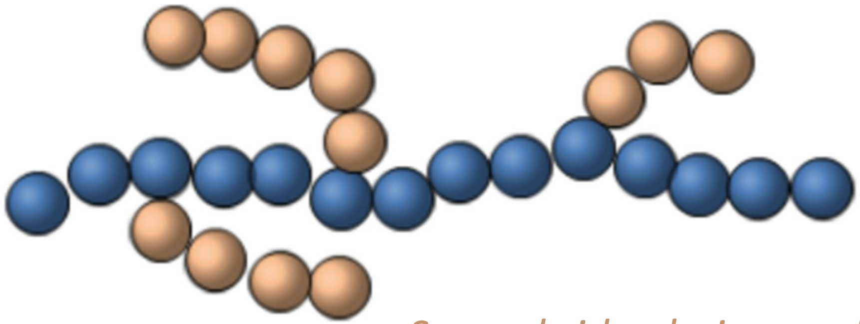
Coupling of Polymer Strands



Graft (Co)Polymers

Structure and properties

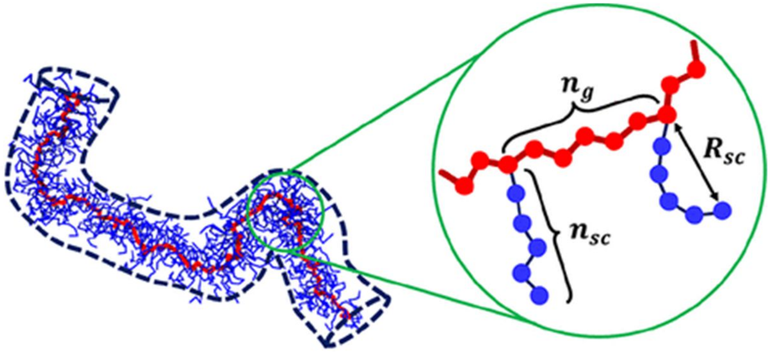
A linear backbone made of one or several monomer(s)

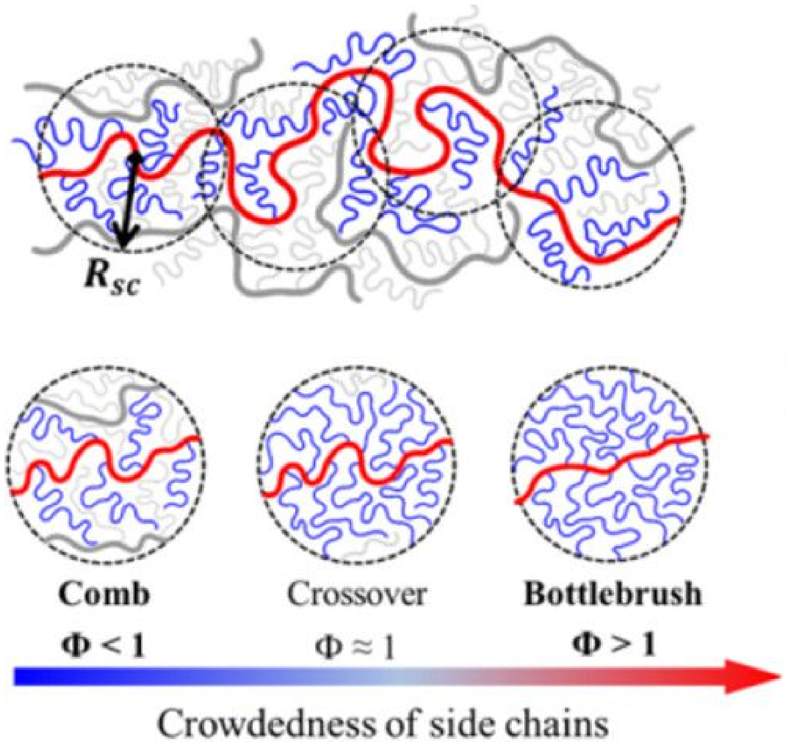


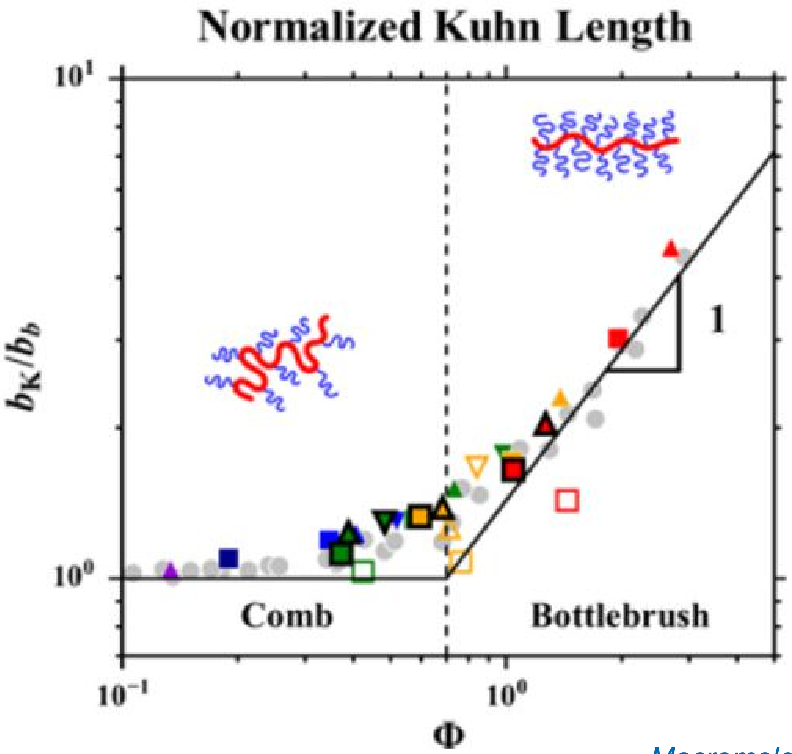
Several side-chains made of repeating units, different or identical to the backbone repeating unit

Graft (Co)Polymers

Comb vs. (Bottle)Brush Polymers



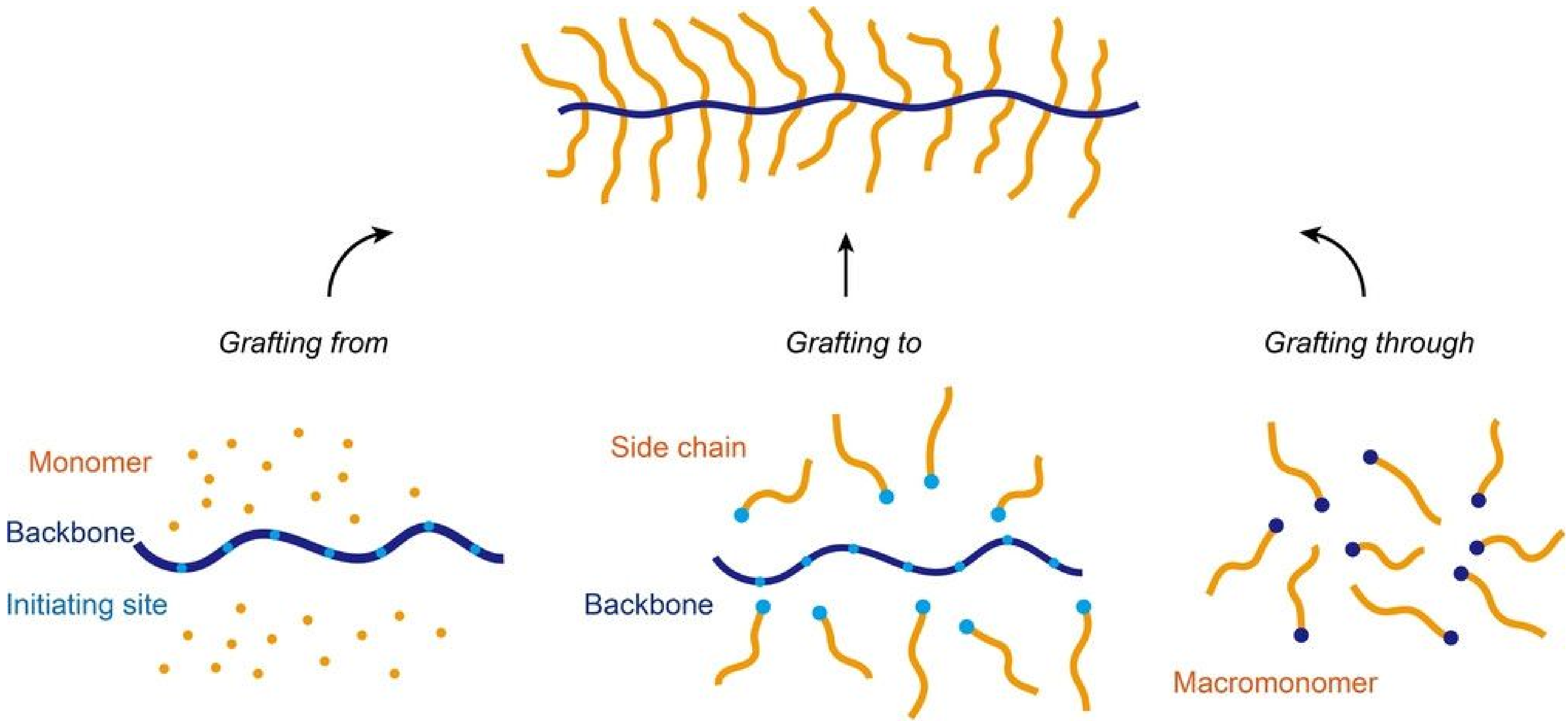




Macromolecules **2019**, *52*, 3942

Graft (Co)Polymers

Synthetic pathways

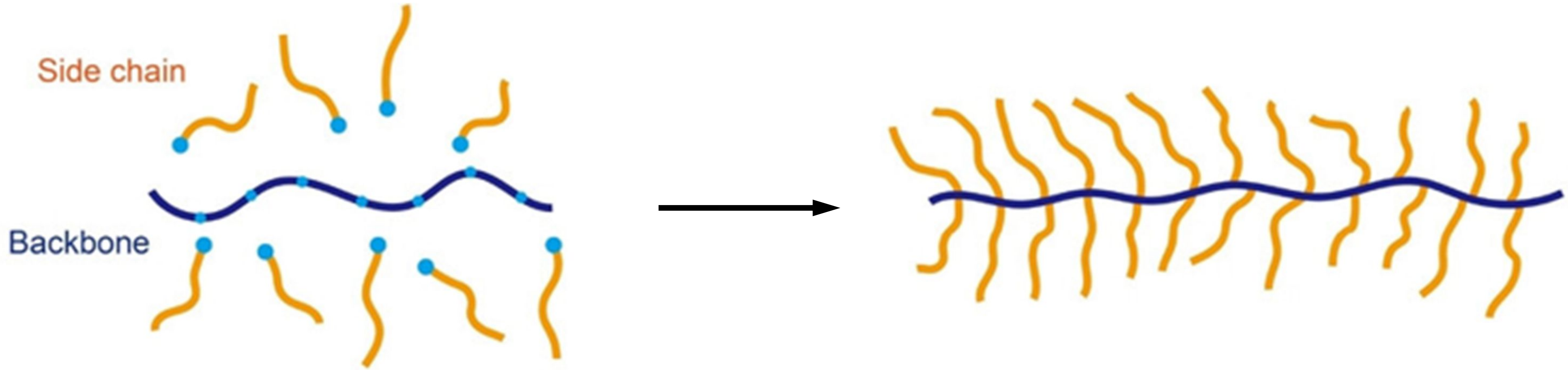


Angew. Chem. Int. Ed. **2023**, *62*, e202219312

Graft (Co)Polymers

Synthetic pathways

Grafting (on)to



Advantages

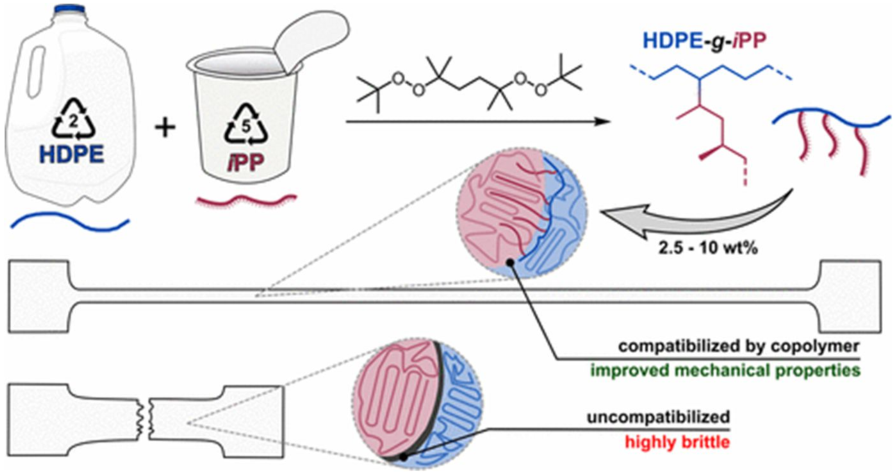
- Well characterized arms;
- Large range of coupling methods;
- No need to incorporate possibly unstable moieties (acrylates, those which are used to polymerize in grafting through)

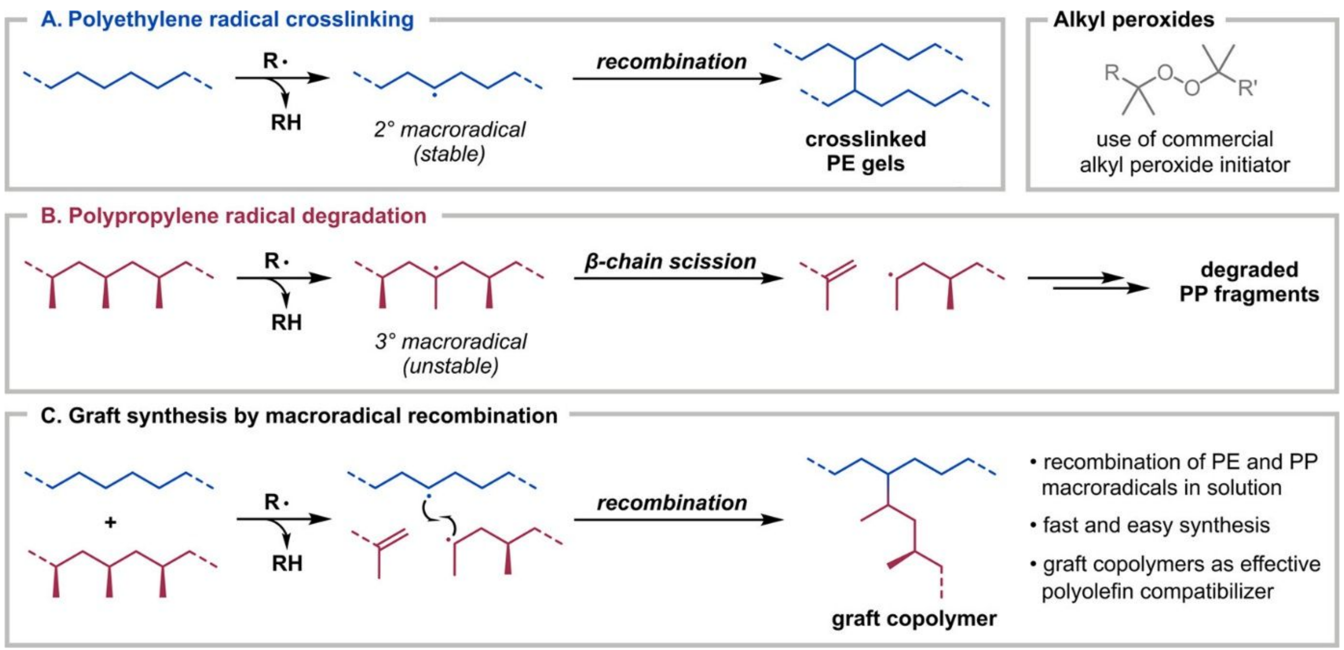
Inconvenients

- Steric hindrance
- Separation grafts from graft polymer

Graft (Co)Polymers

Synthetic pathways



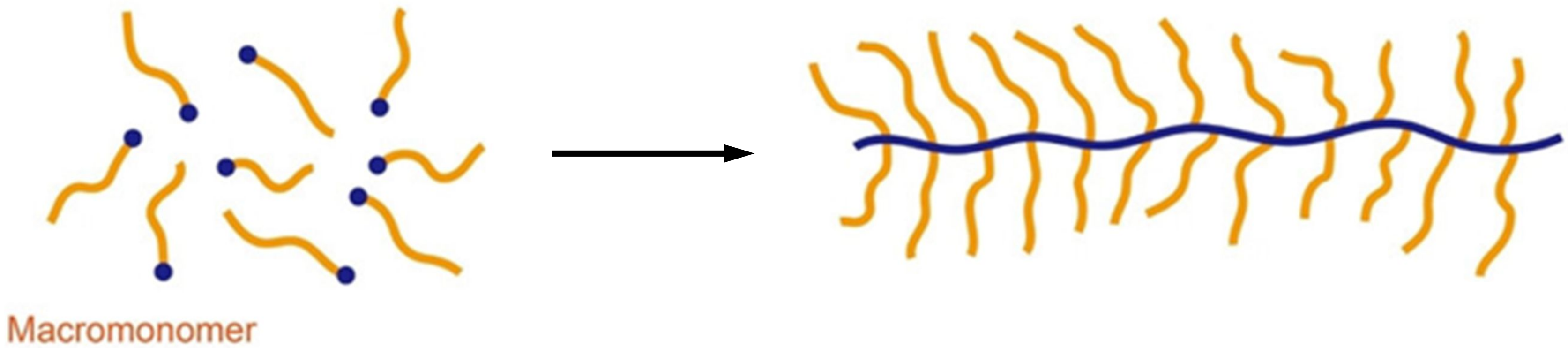


J. Am. Chem. Soc. **2025**, *147*, 19052

Graft (Co)Polymers

Synthetic pathways

Grafting through



Advantages

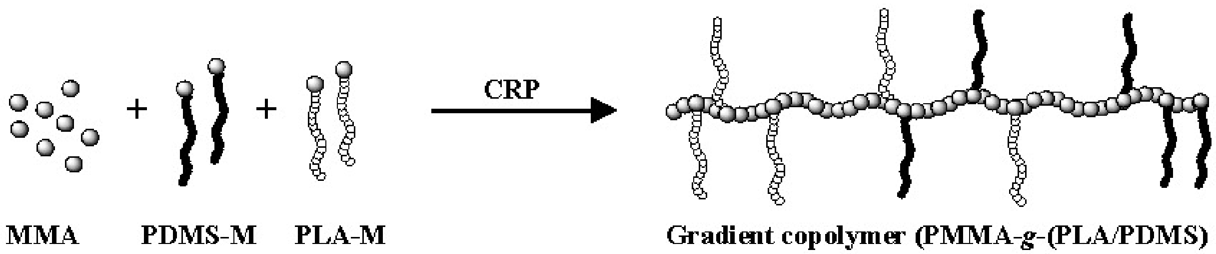
- Well characterized arms;
- Usually highly efficient chemistry (the one used for propagation)
- Access to well-defined graft polymers with different graft types

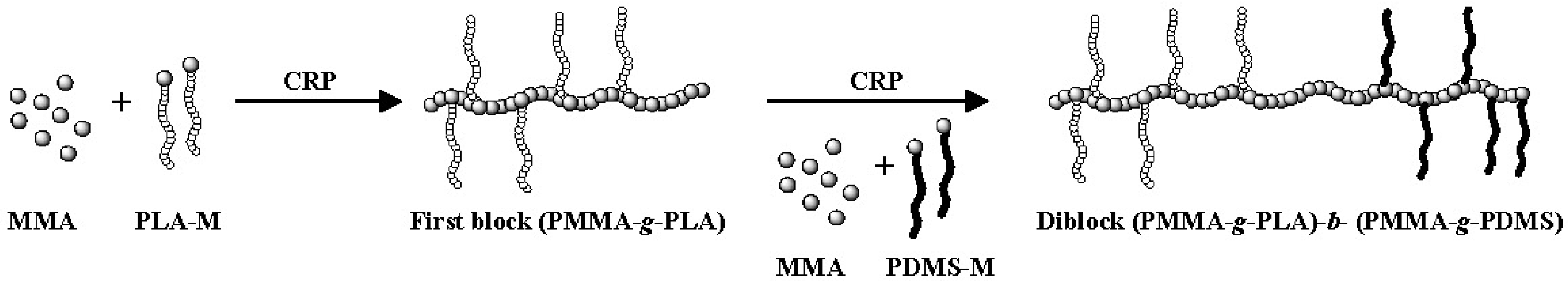
Inconvenients

- Steric hindrance
- Separation grafts from graft polymer (but potentially less side-products than in *grafting (on)to*)

Graft (Co)Polymers

Synthetic pathways

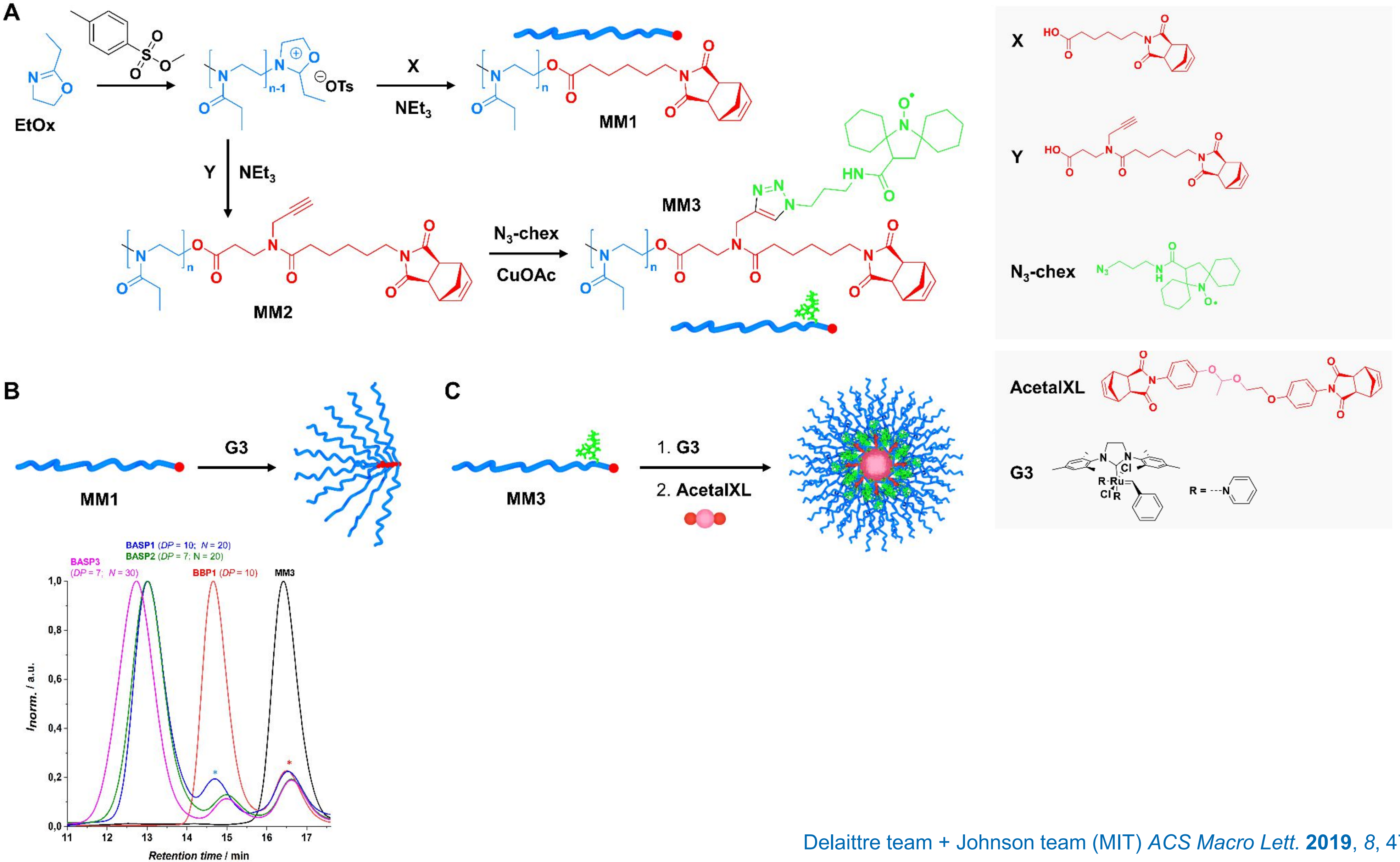




J. Polym. Sci., Part A: Polym. Chem. **2004**, *42*, 1939

Graft (Co)Polymers

Synthetic pathways

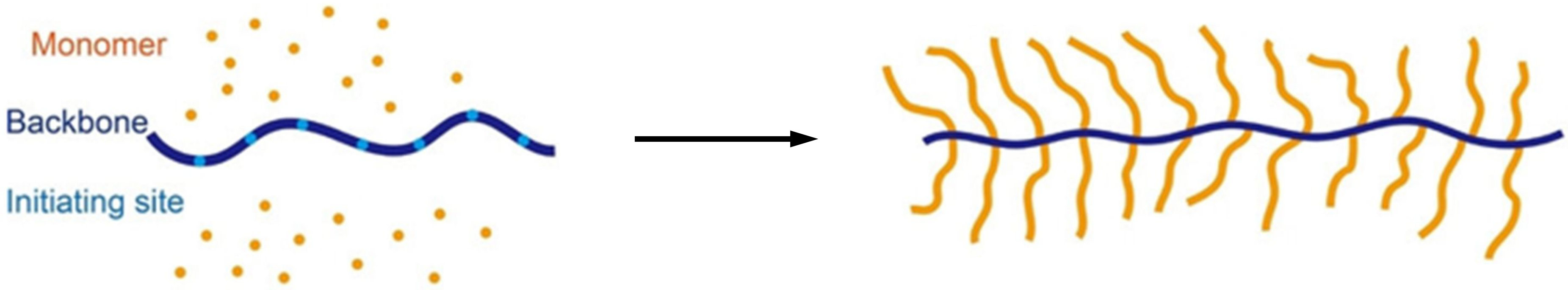


Delaittre team + Johnson team (MIT) *ACS Macro Lett.* **2019**, *8*, 473

Graft (Co)Polymers

Synthetic pathways

Grafting from

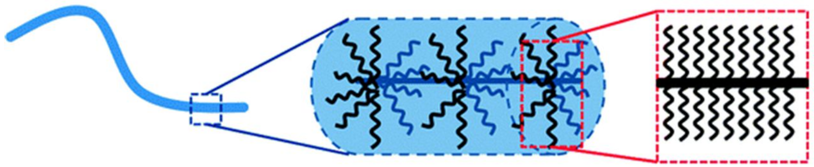


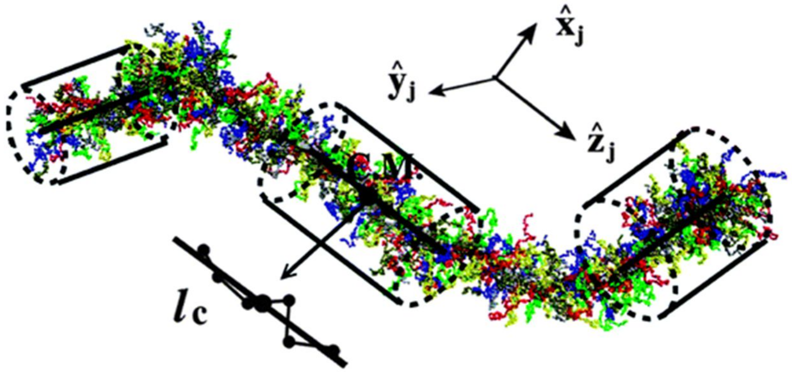
Advantages
- No steric hindrance
- Easy separation between monomer and graft polymer

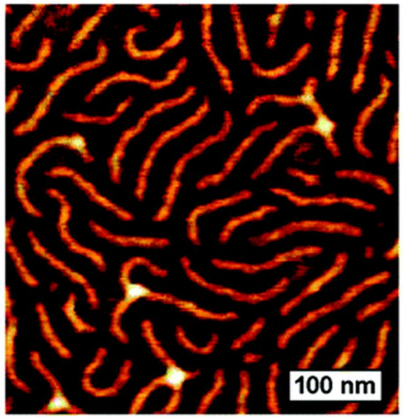
Inconvenients
- Possible termination between growing grafts
- Arms are not well-characterized
- Depending on transfer reactions, possible contamination by homopolymers

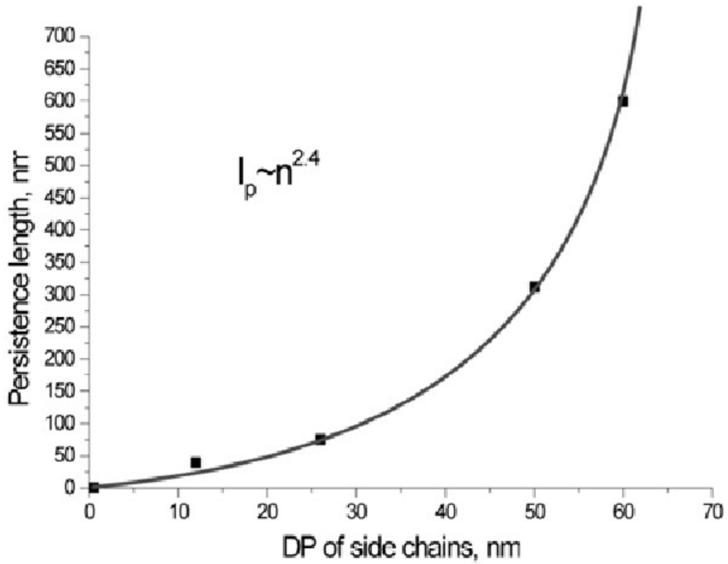
Graft (Co)Polymers

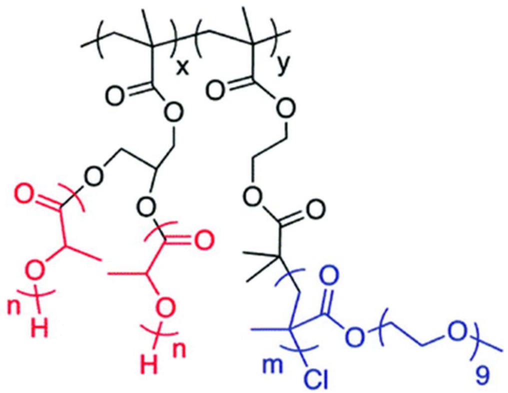
(Bottle)Brush Polymers







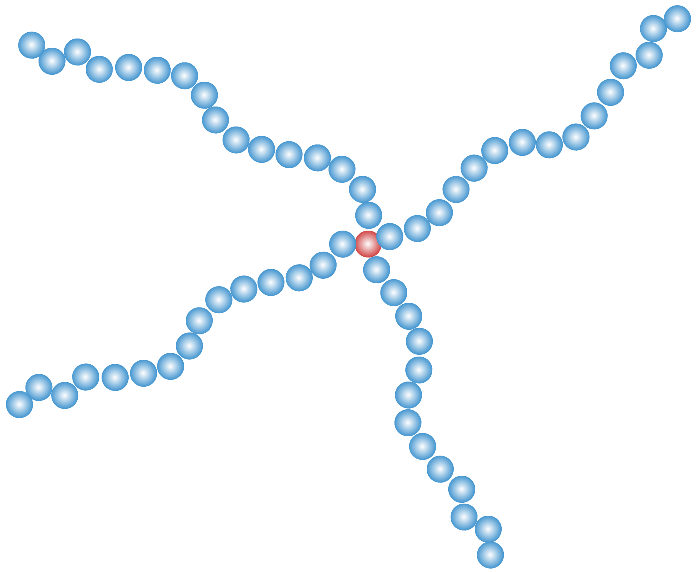




Chem. Soc. Rev. **2015**, *44*, 2405

Star (Co)Polymers

Structure and properties

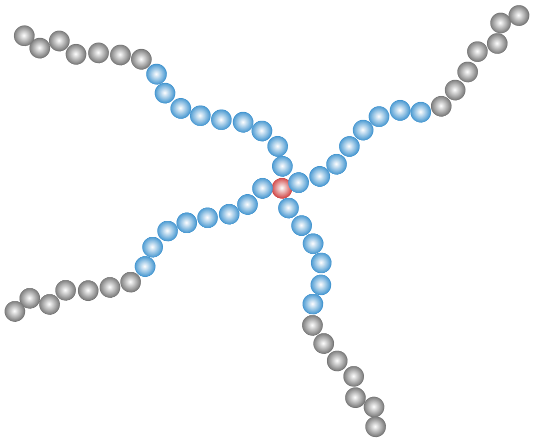


A graft polymer, where all grafts are connected to a single point/location, not along a linear chain anymore

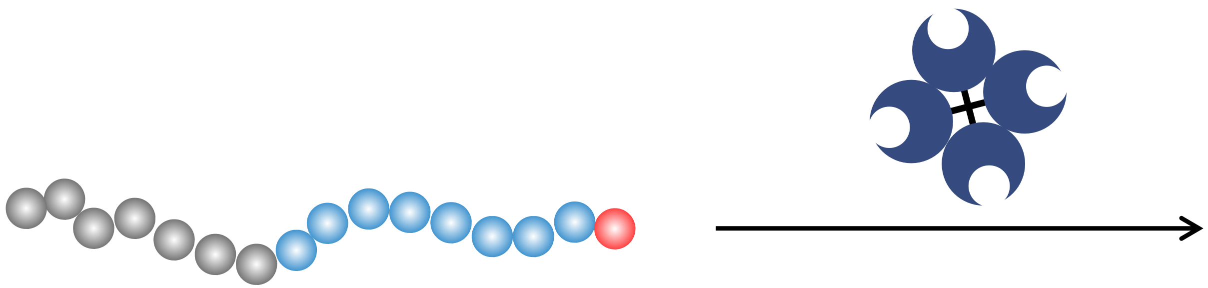
Often necessary: a multifunctional core molecule.
Typical examples are trimethylolpropane, glycerol, erythritol, as well as macrocycles such as cyclodextrins and porphyrins,

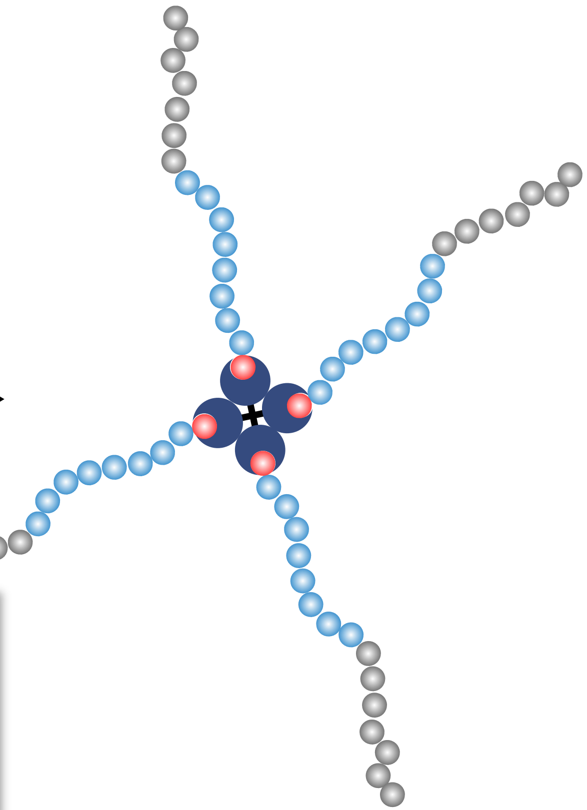
Star (Co)Polymers

Synthetic pathways



Arm First



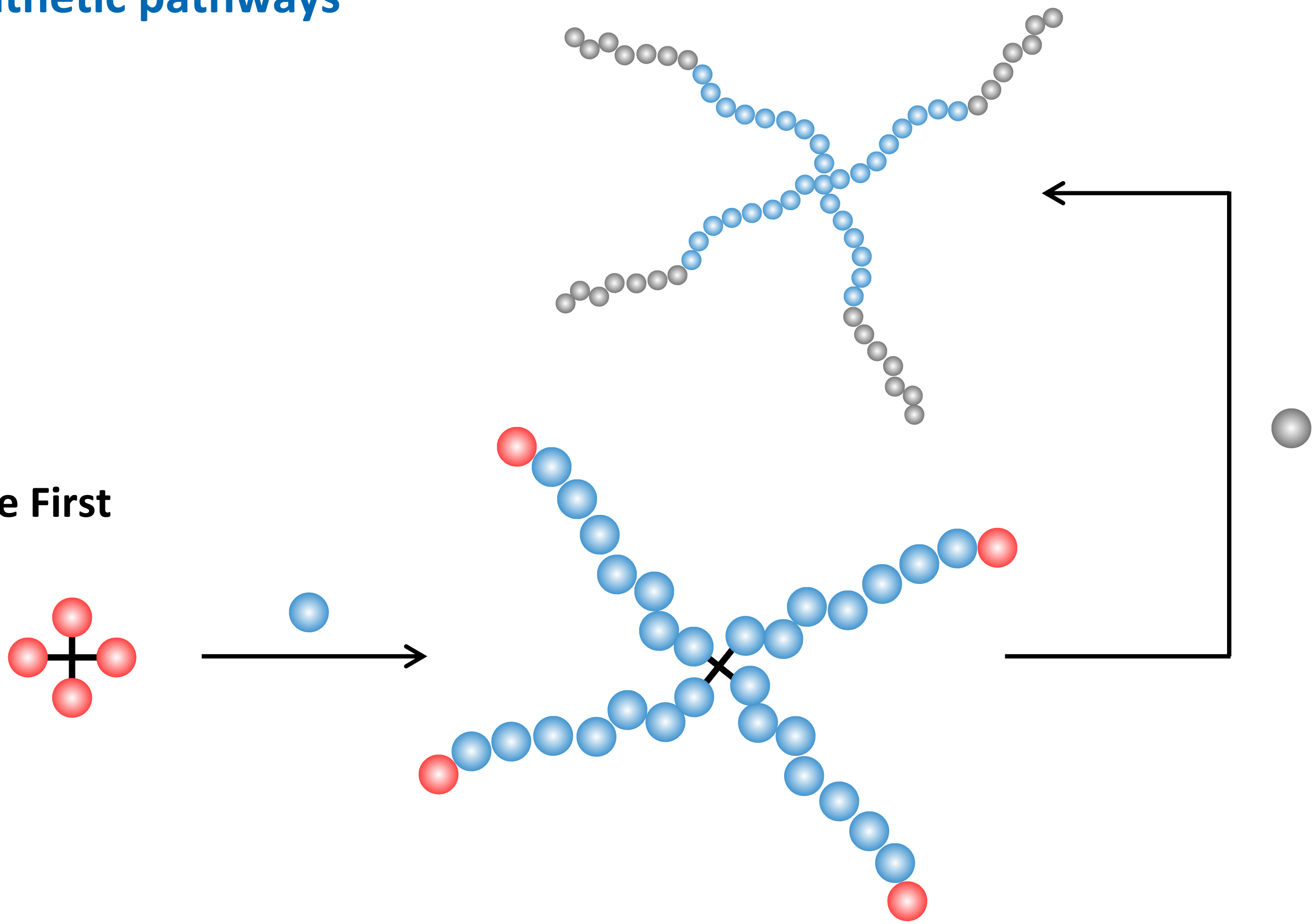


Properties of star polymers
(in comparison to linear analogues
of same mass and composition)
→ Lower R_h, R_g, viscosity → Self-assembly properties

Star (Co)Polymers

Synthetic pathways

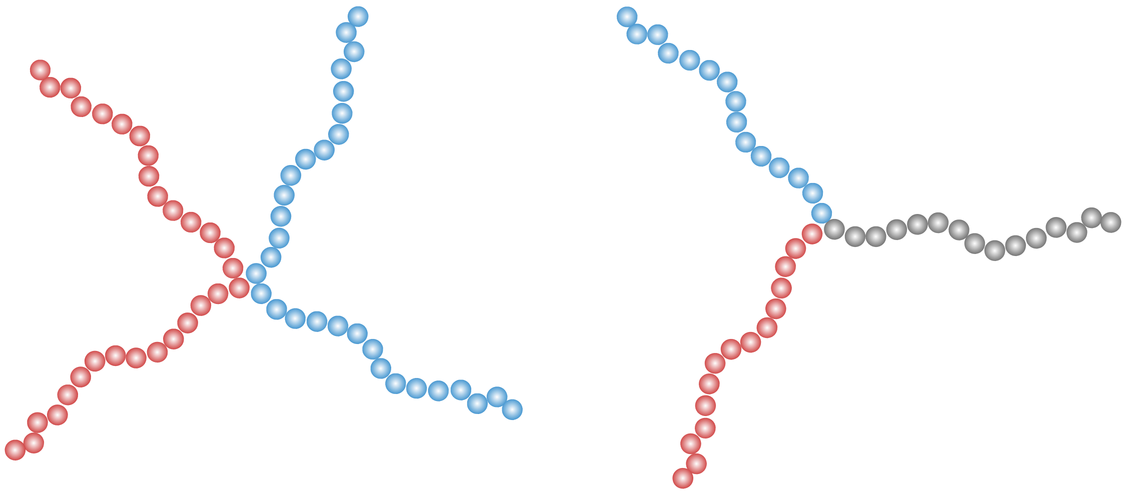
Core First



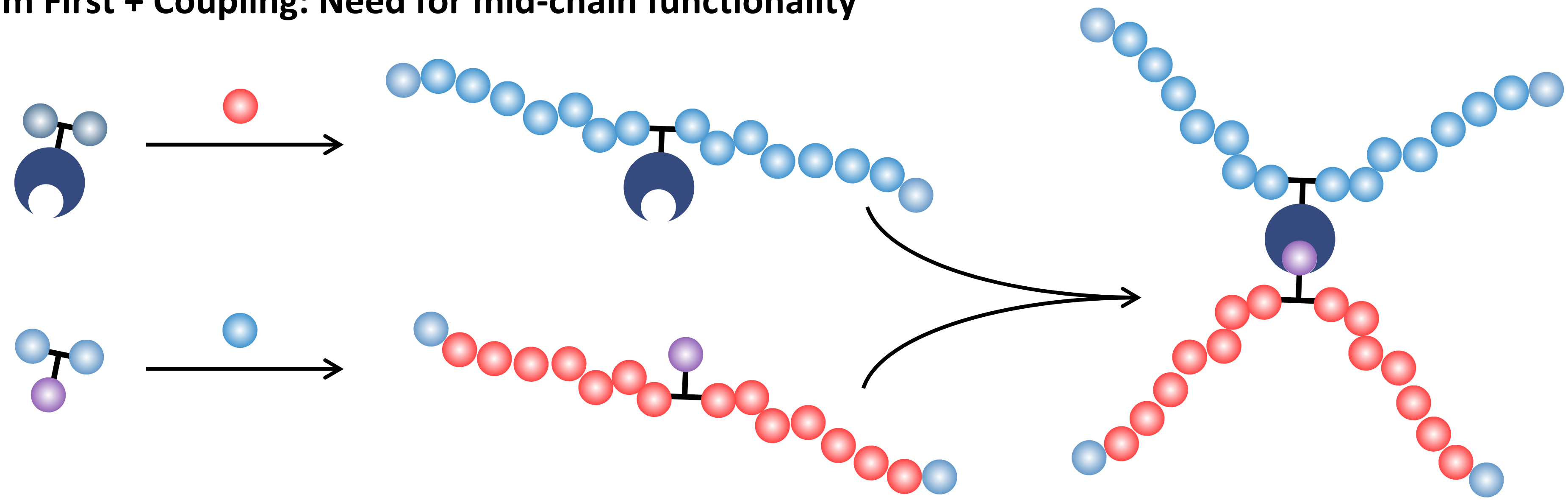
Obviously, also possible: Combination of both methods

Star (Co)Polymers

Synthetic pathways

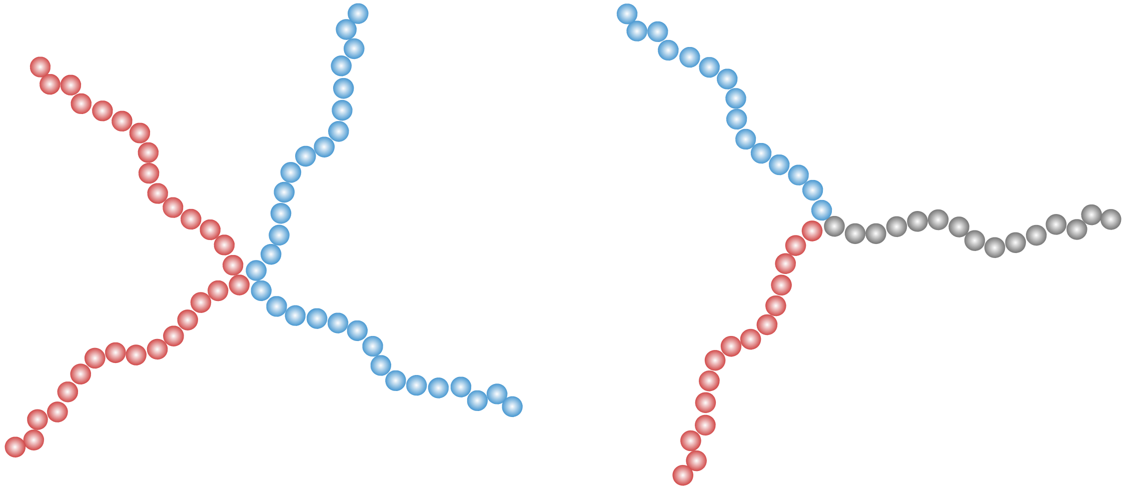


Arm First + Coupling: Need for mid-chain functionality

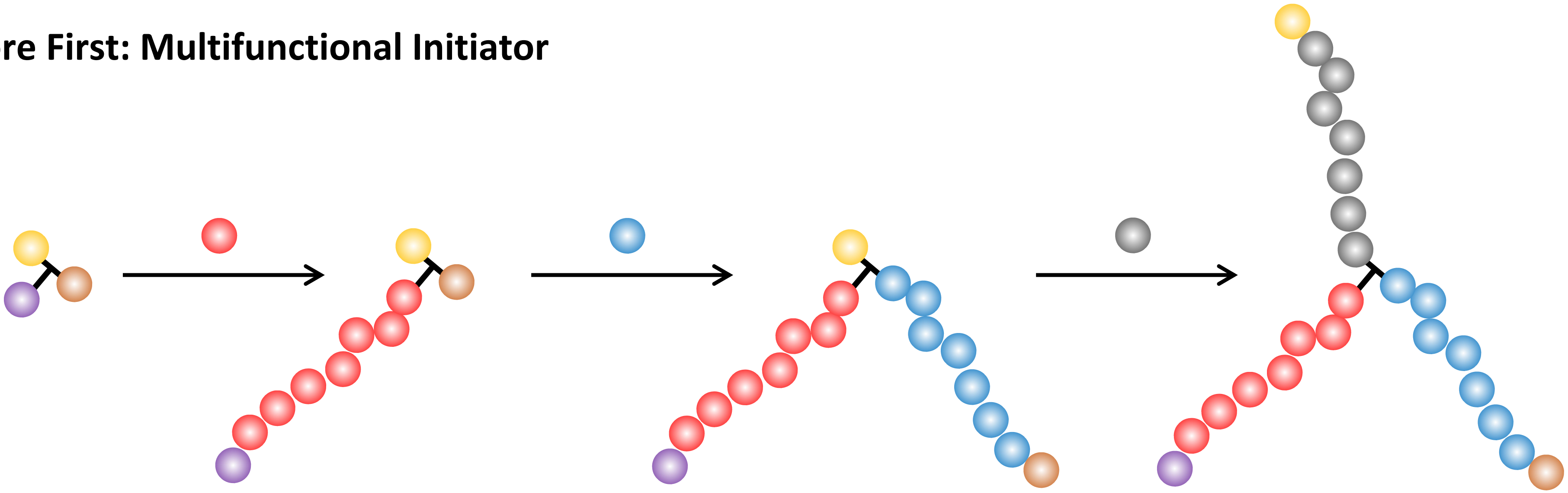


Star (Co)Polymers

Synthetic pathways

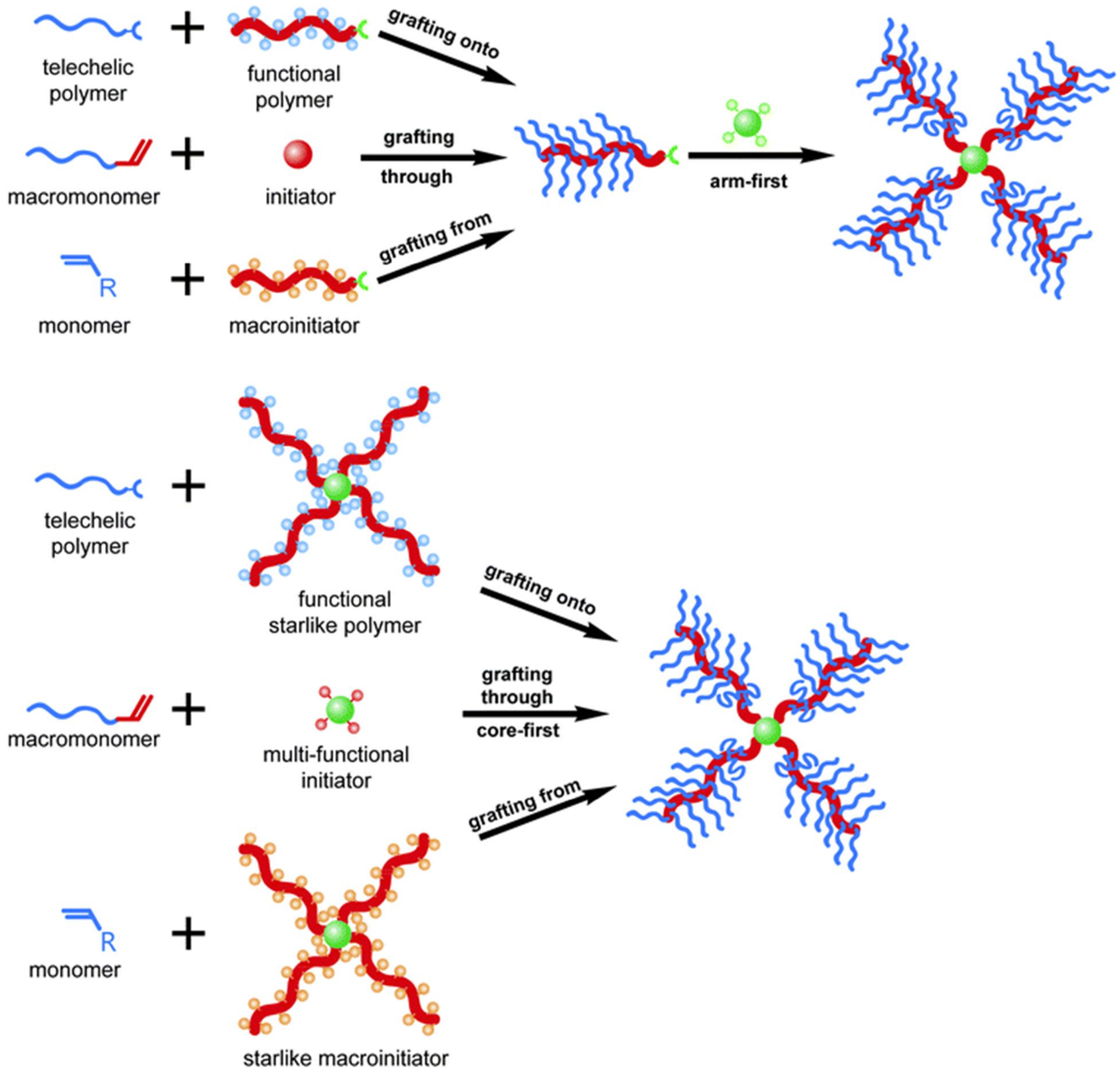


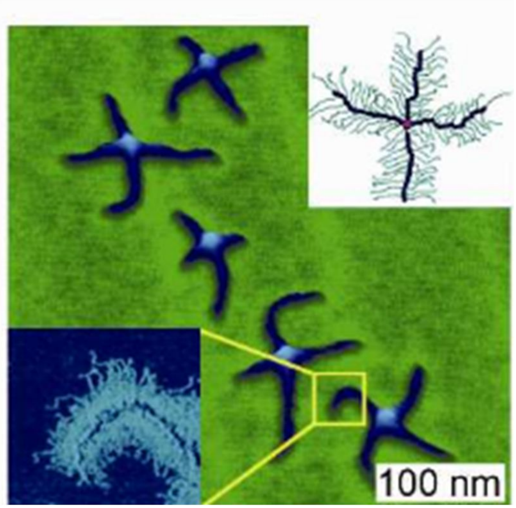
Core First: Multifunctional Initiator



Star Graft Copolymers

Merging to the two types of architectures





Polym. Chem. **2013**, *4*, 1289